DTSC 545 Prompt Engineering

Module 7 - Final Project Instructions

Final Project Instructions

Assignment Overview
In this final project, you will demonstrate your mastery of prompt engineering by designing and refining a prompt-based AI solution for a real or hypothetical business challenge. This project serves as the culmination of your learning, allowing you to apply technical, ethical, and evaluative skills to a practical scenario.

Detailed Instructions:
1. Select a Business Challenge
- Identify a real or hypothetical business problem that can be addressed through prompt engineering.
- Your challenge should be specific, measurable, and relevant to a real-world business context (for example: automating HR feedback summaries, generating financial forecasts, or creating marketing copy).

2. Design and Refine Prompts
- Create three (3) distinct prompts designed to solve different aspects of the identified challenge.
- Demonstrate at least four (4) prompting techniques (for example: zero-shot, few-shot, chain-of-thought, role prompting, or self-reflection).
- Show at least one refinement cycle where you improved a prompt based on model feedback or evaluation results.

3. Evaluate Prompt Effectiveness
- Apply evaluation metrics such as accuracy, completeness, tone, and relevance.
- Include a short rubric (can be a simple table or checklist) describing how you assessed each prompt, and include a numerical score to each (as was done in the lecture videos).

4. Develop Deployment Recommendations
- Suggest how your prompt-based solution could be integrated into a business workflow.
- Address considerations for scalability, bias mitigation, and governance to ensure responsible use.

Deliverables
Your final report should be organized into four clearly labeled sections corresponding to the project instructions:

1. Section 1 – Business Challenge Description
Describe the real or hypothetical business challenge you selected. Explain why it matters to the organization and what outcomes the AI solution aims to improve.

2. Section 2 – Prompt Design, Outputs, and Techniques Used
Design and test three (3) distinct prompts that each address a different aspect of your chosen business challenge. Together, these prompts should demonstrate how prompt engineering can support multiple stages or perspectives of solving the same problem (for example: generating summaries, extracting data, analyzing patterns, or formatting outputs).

For each prompt:

- Include the prompt text and at least one example output.
- Identify which prompting technique(s) were used (e.g., zero-shot, few-shot, chain-of-thought, role prompting, or self-reflection).

After creating all three prompts, select one (1) to refine further.

- Show an iterative improvement cycle where you revise the prompt based on performance or evaluation feedback.
- Clearly explain what changes you made and how those changes improved accuracy, tone, or structure.

3. Section 3 – Evaluation and Metrics
Provide a short rubric or table outlining your evaluation criteria (such as accuracy, completeness, tone, relevance, and bias awareness), including a numerical score (as was done in the lecture videos). Discuss how each prompt performed against your criteria and summarize key findings or insights from testing.

4. Section 4 – Business-Ready AI Prompting Solution and Recommendations
Propose how your prompt-based solution could be implemented within a business workflow. Justify your recommendations with respect to scalability, governance, and ethical considerations. Conclude with lessons learned and next-step opportunities for improvement or deployment.

Submission Guidelines:
Upload a single, clearly labeled, pdf file containing all four sections to Brightspace. Please label the file as "lastname_firstname_project". The report, including the prompt inputs and outputs, should not exceed 12 pages long, using size 11 or 12 font. If you require additional pages for structured prompt outputs (i.e. code, tables, excel sheets etc.), please include an appendix.

Due Date:

The final project is due at the end of the term. If you submit it by the end of Week 5, you will earn 5% extra credit, with your final grade capped at 100%.

Final Project Rubric

Criteria	Exemplary (25–23 pts)	Proficient (22–19 pts)	Developing (18–15 pts)	Needs Improvement (14–0 pts)
1. Business Challenge Description (25 pts)	Clearly defines a realistic and meaningful business challenge. Provides rich context and demonstrates a strong understanding of the organizational impact and desired outcomes.	Defines a relevant business challenge with sufficient context. Explains the problem and goals with moderate depth.	Challenge is vague or lacks detail. Minimal explanation of context or business importance.	Challenge is unclear, irrelevant, or not connected to course objectives.
2. Prompt Design, Outputs, and Techniques Used (25 pts)	Presents three well-structured, clearly written prompts with accurate outputs. Demonstrates at least four prompting techniques effectively and shows evidence of thoughtful refinement.	Presents three prompts with mostly accurate outputs. Demonstrates several prompting techniques and some refinement effort.	Presents fewer than three complete prompts or inconsistent outputs. Techniques are limited or not clearly explained.	Prompts are missing, incomplete, or show no application of prompting techniques.
3. Evaluation and Metrics (25 pts)	Provides a clear, well-designed evaluation rubric with multiple relevant criteria. Demonstrates deep analysis of results, including reflection on performance, bias, and improvement.	Provides a functional evaluation rubric with some relevant metrics. Includes analysis of prompt performance with basic interpretation.	Evaluation rubric lacks structure or detail. Analysis is brief or missing key performance insights.	Evaluation is missing or does not meaningfully assess prompt effectiveness.
4. Business-Ready AI Solution and Recommendations (25 pts)	Presents a polished, professional solution with strong justification for implementation. Considers scalability, governance, and ethical issues with thoughtful recommendations.	Presents a clear and feasible solution with reasonable justification. Mentions governance and ethics considerations.	Solution is partially developed or lacks clear justification. Limited discussion of governance or ethics.	Solution and recommendations are missing, unclear, or lack practical or ethical consideration.